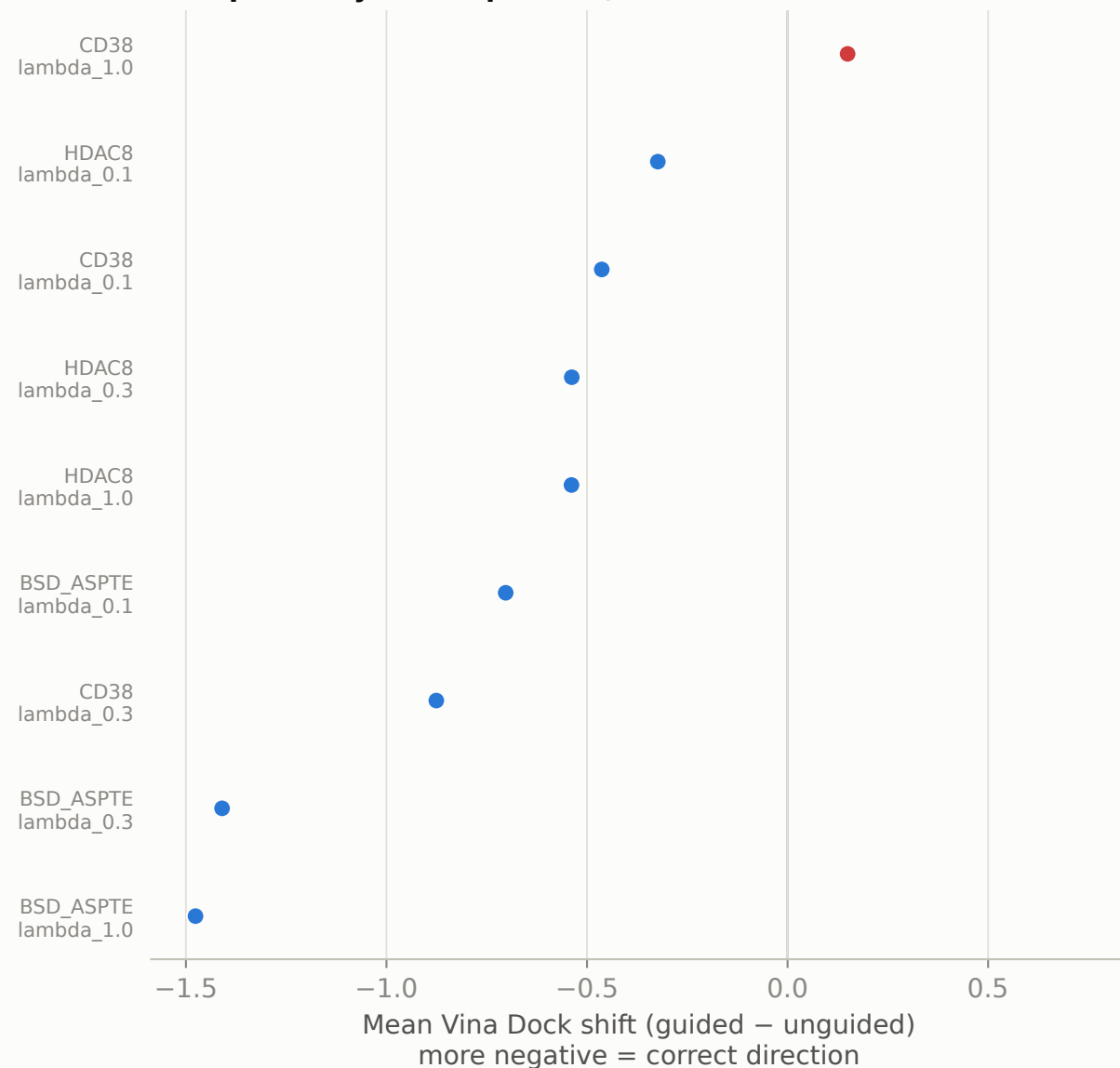
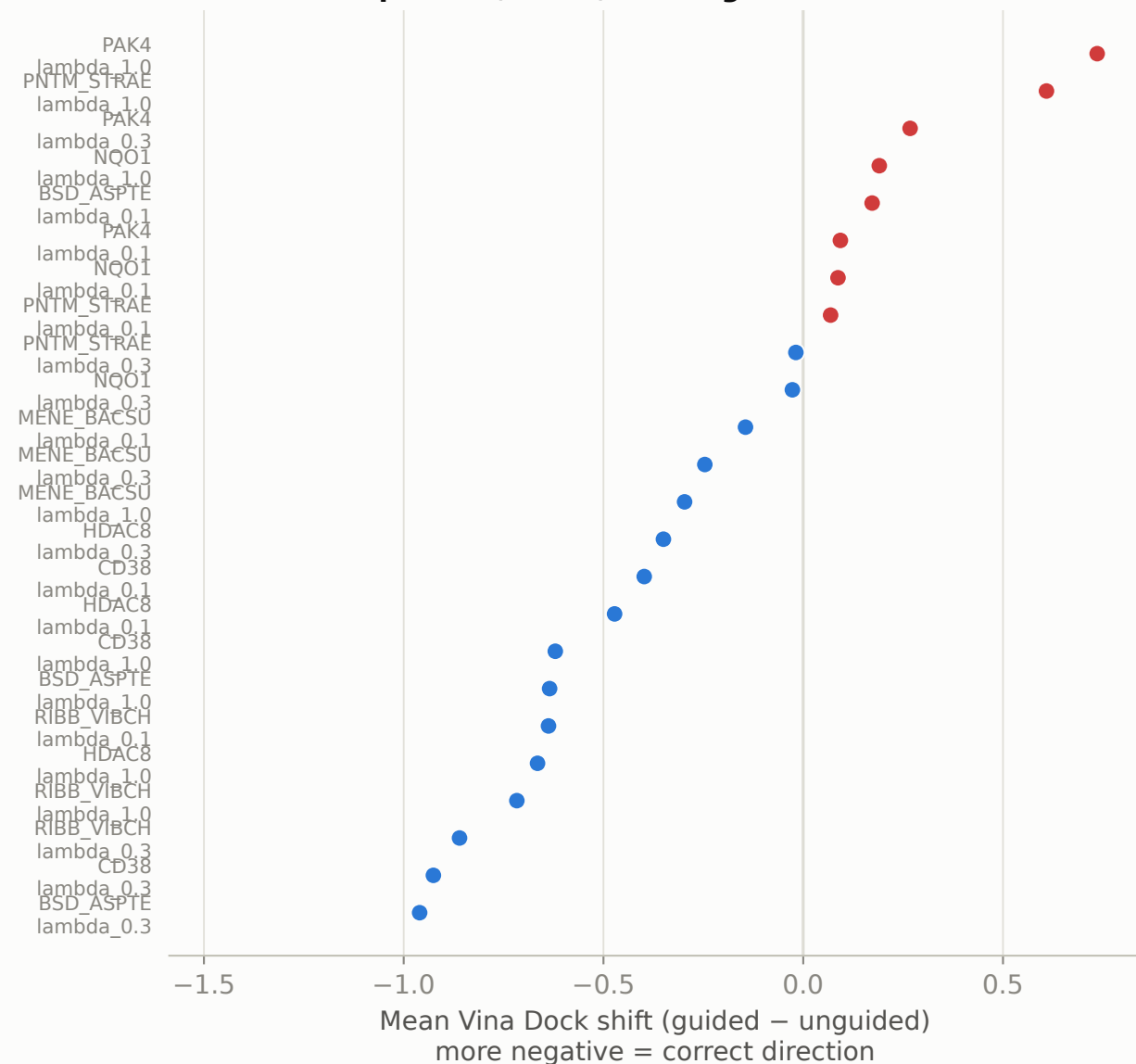


# Track A: physics-anchored gradient guidance (GIGN+PIGNet2) — direction consistency weakens, not strengthens, with more data and full convergence

## A. Exploratory tier (3 pockets, n=20)



## B. Full tier (8 pockets, n=30, converged model)



● Correct direction (n.s.)
 ● Wrong direction (n.s.)
 ● Significant after BH correction